Kernix 0.11.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

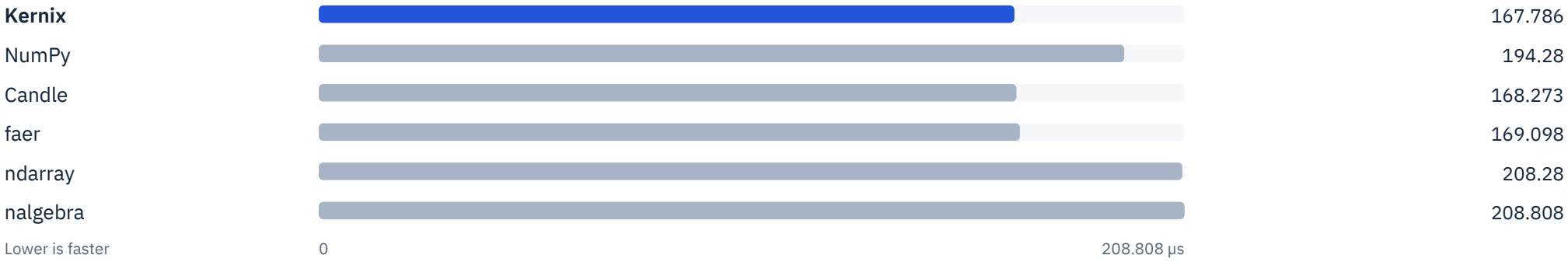
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



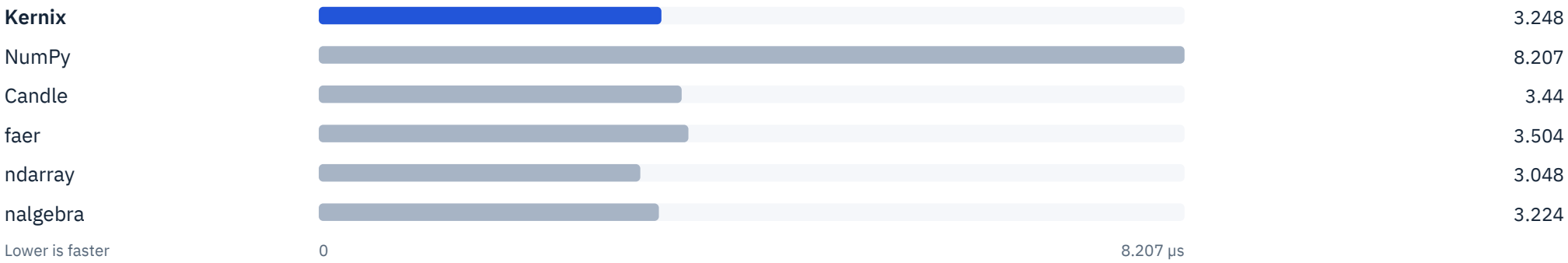
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

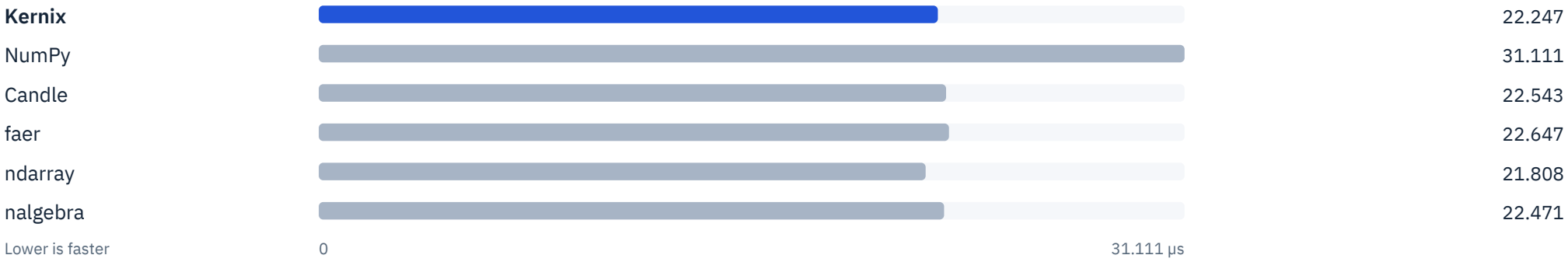
32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



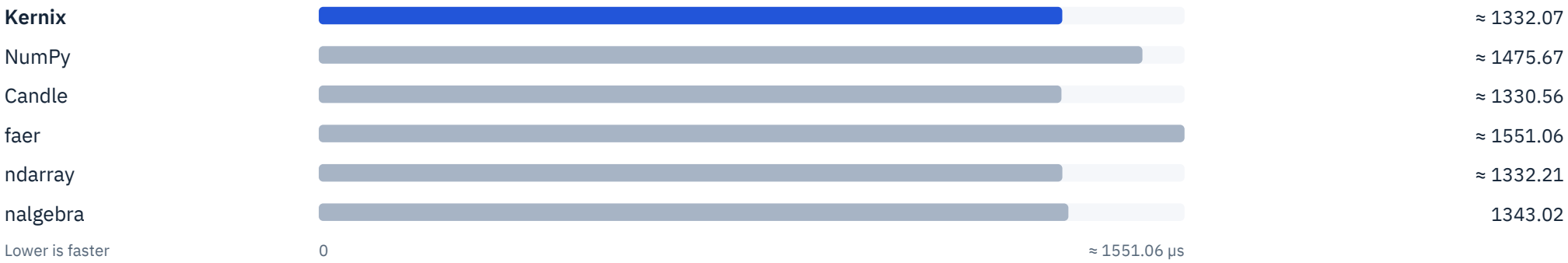
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

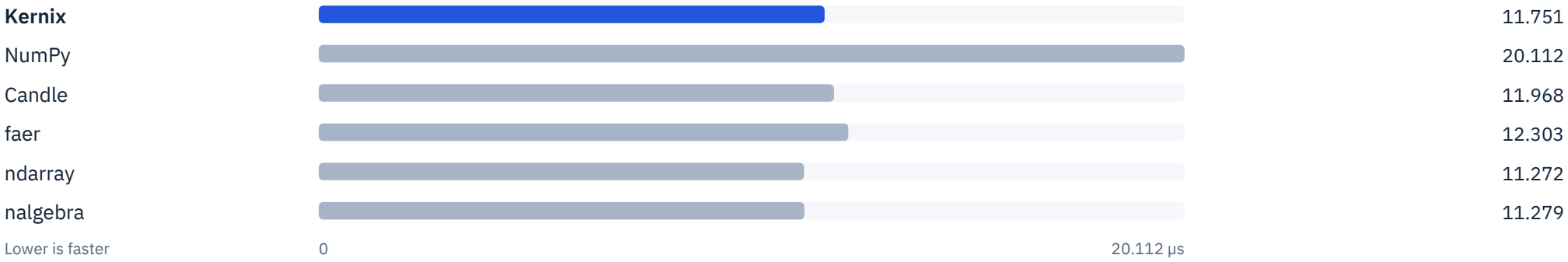
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



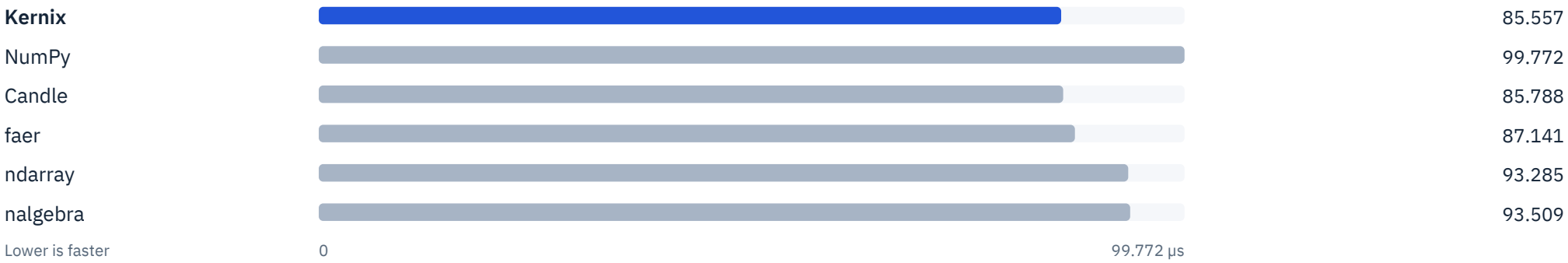
Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

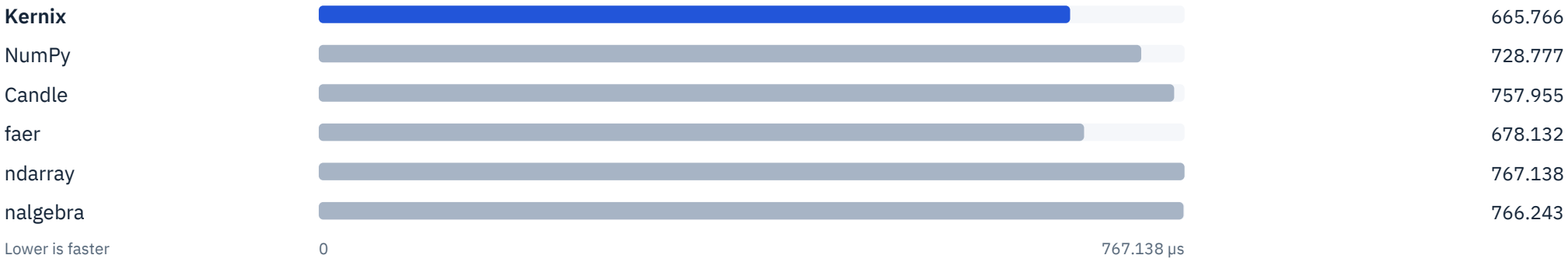
128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



Kernix 0.11.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.11.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.11.0 / Explicit parallel execution

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



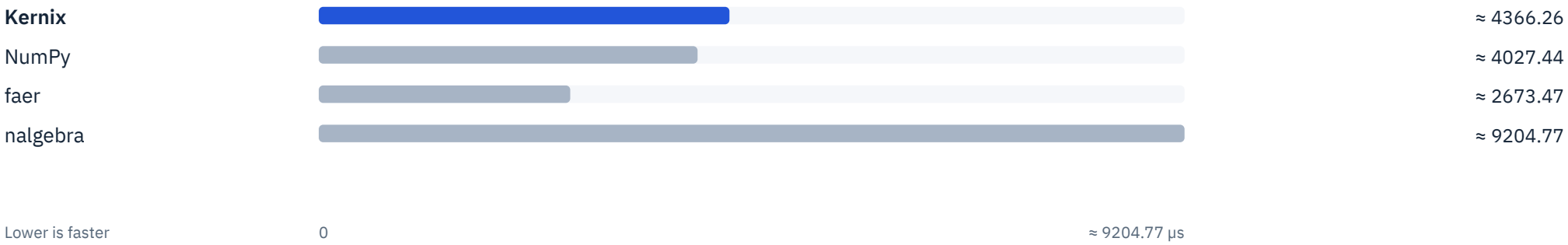
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

512 × 512 · f64 · 1 worker · allocated output

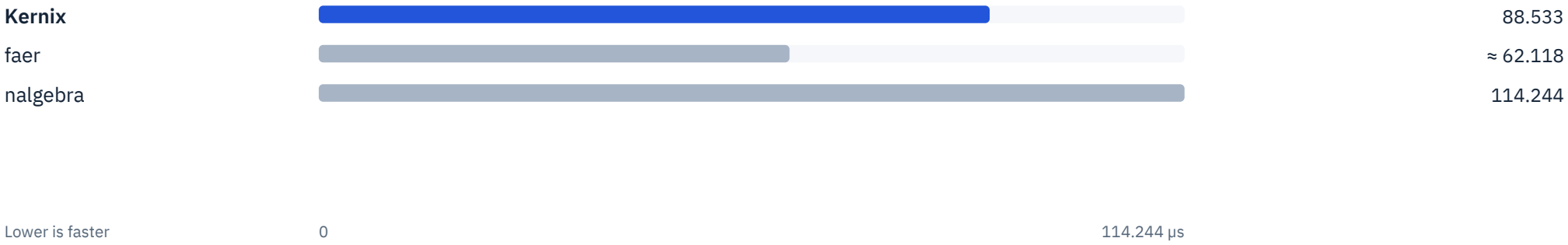
Per-call median



Column-pivoted QR

96 × 32 · f64 · 1 worker · allocated output

Per-call median



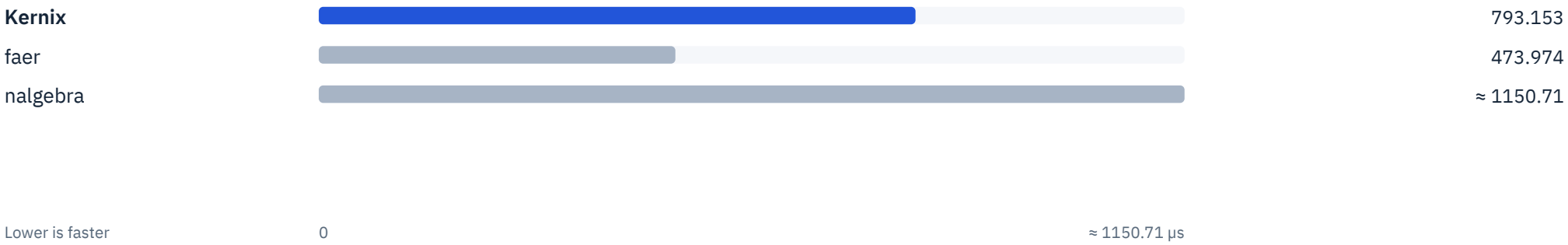
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

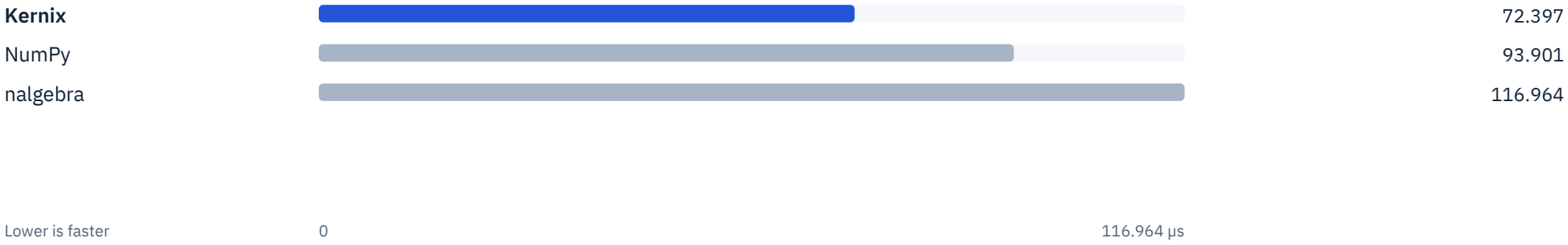
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



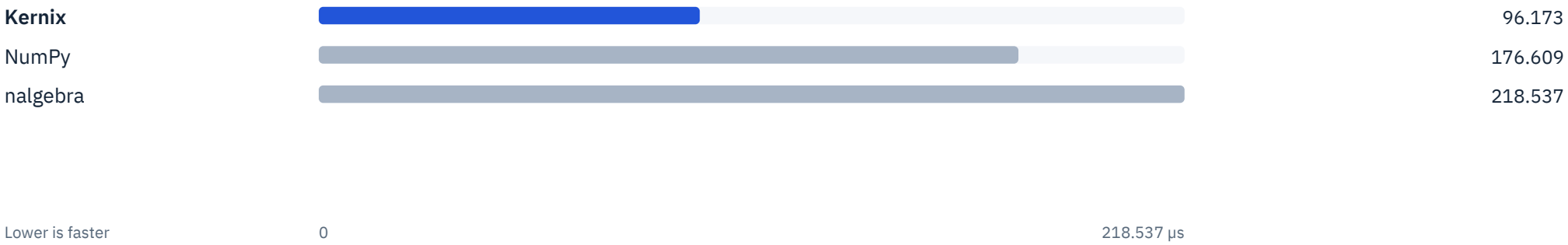
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Minimum-norm least squares

Per-call median

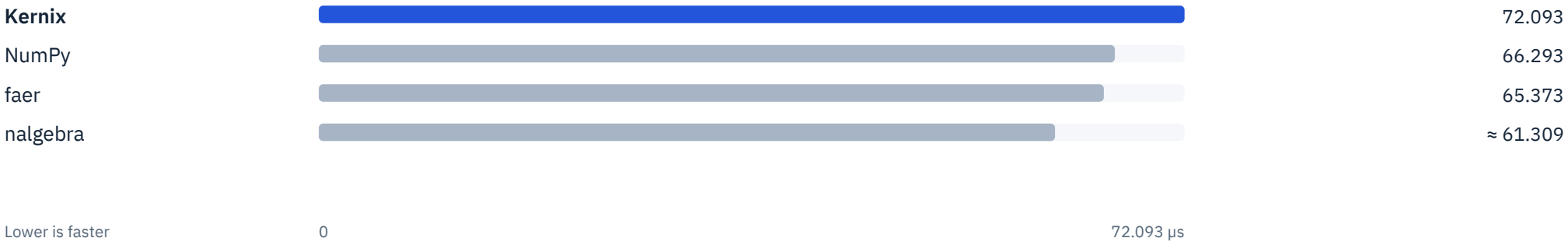
32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Thin QR

256 × 64 · f64 · 1 worker · allocated output

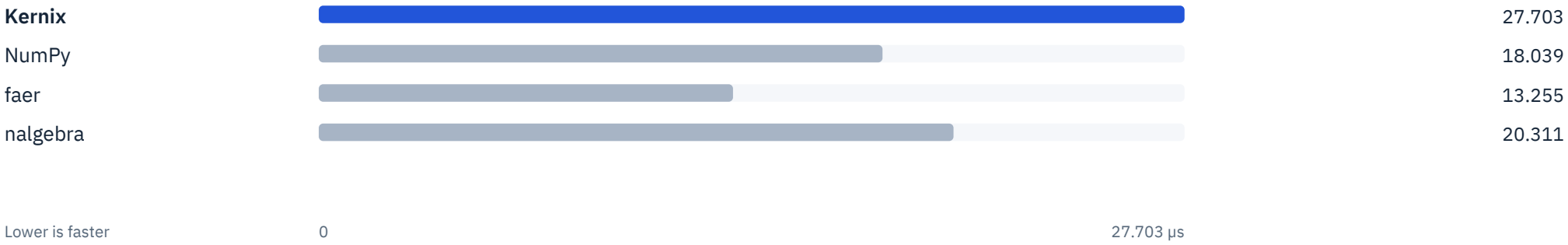
Per-call median



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output

Per-call median



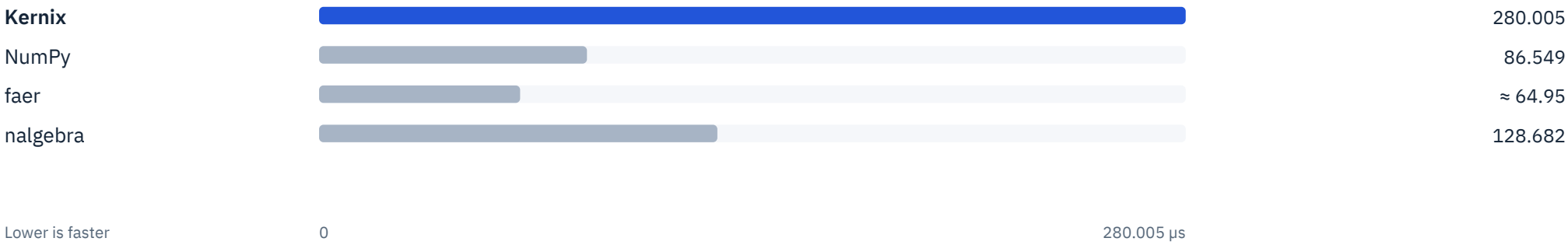
Kernix 0.11.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output

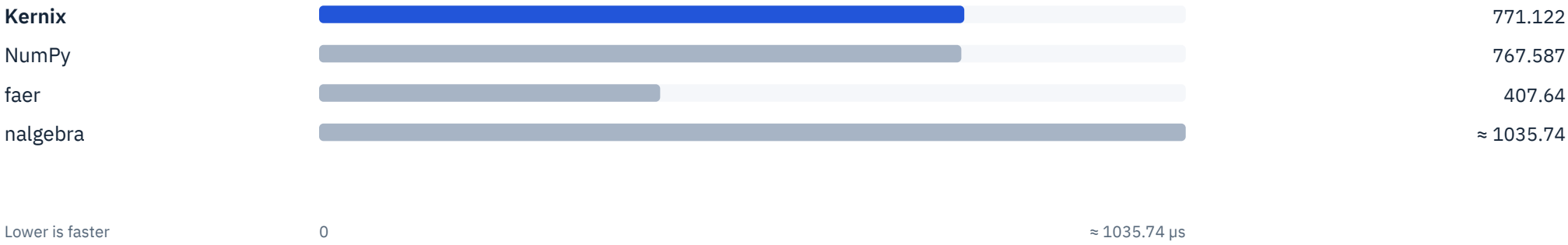
Per-call median



Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



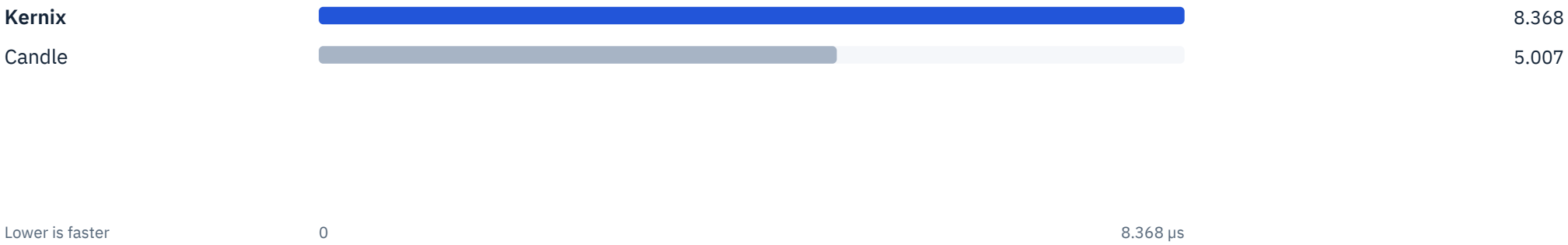
Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product

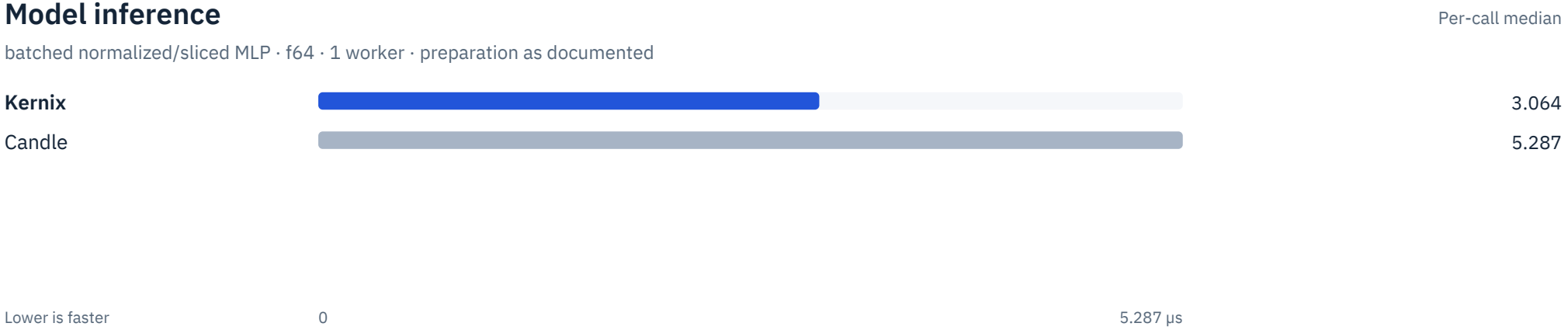


Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

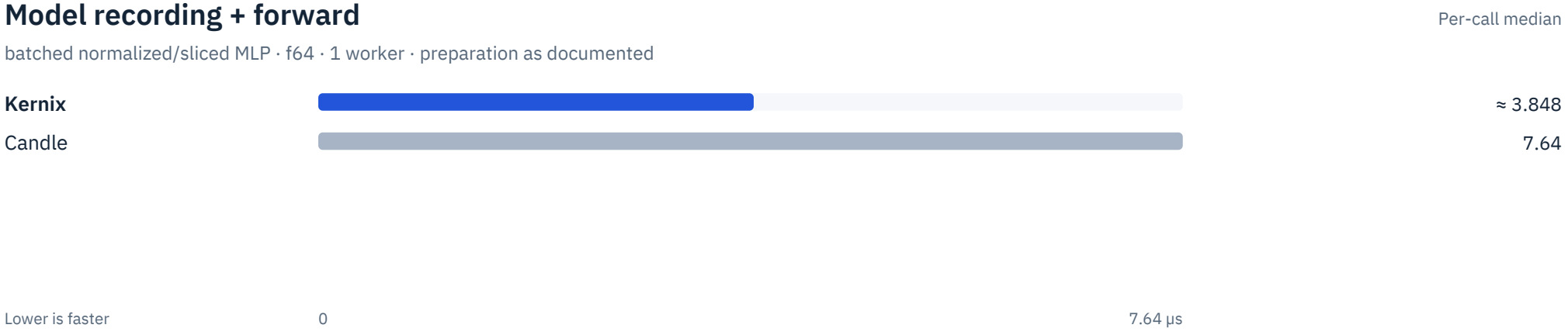
Model inference

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.11.0 / Tensor contractions

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

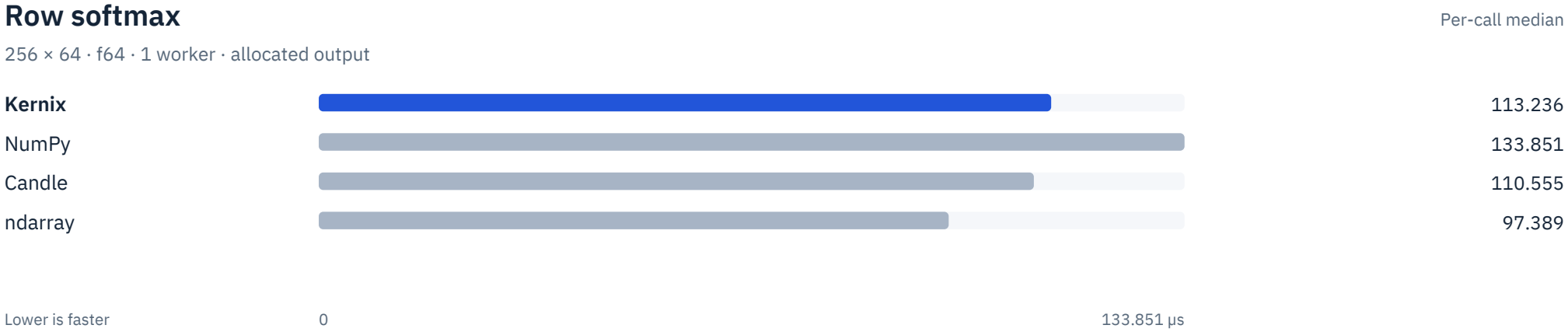


Kernix 0.11.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



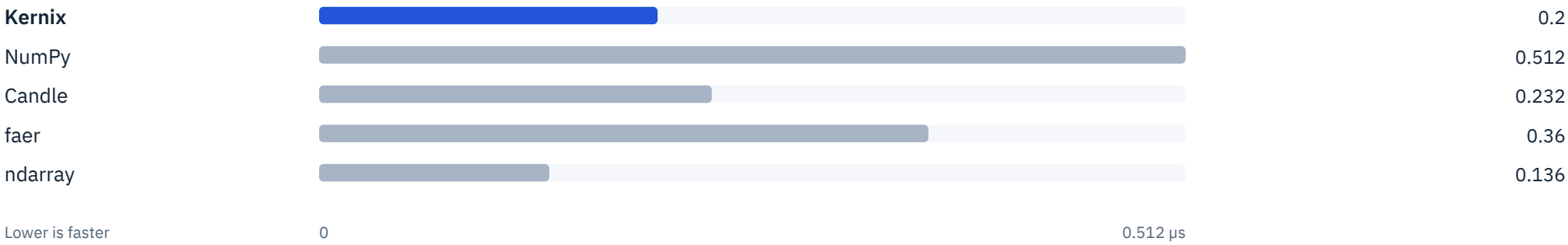
Kernix 0.11.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8039757672973089	0.7961028439858441 – 0.8075187028214404

Matrix multiplication

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9104404550339998	0.9083013425523341 – 0.9116032637490727

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matrix multiplication

256 × 256 · 256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9937348734791259	0.9925390619170559 – 0.9968513537905461

Matrix multiplication

256 × 256 · 256 × 256 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8616485177090156	0.8596499666554435 – 0.8637385121301779

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Independent matrix operands

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.7968843618933493	0.7850777114855892 – 0.8007173985605017

Independent matrix operands

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9059853428794695	0.9014599690978463 – 0.9103167103740432

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

128 × 128 · f64

Observed time increase



CURRENT / BASELINE	95% INTERVAL
1.00357375150991	1.0008964317729707 – 1.0050536592369261

Cholesky factorization

256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.3869933969656744	0.3851419972722835 – 0.38939434067885376

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Cholesky factorization

512 × 512 · f64

Observed time reduction



CURRENT / BASELINE	0.2852289637520951	95% INTERVAL	0.2845246063012664 – 0.2860102284176894

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f64

Observed time reduction



CURRENT / BASELINE	0.9447061485610845	95% INTERVAL	0.9284550919482365 – 0.9752992394224139

Version comparison / paired processes

Neoverse-N2 · 9 process pairs per workload · ratio below 1 means less time

Matmul · 2 workers · reused output

256 × 256 · 256 × 256 · f64

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.8347739732517124	0.8272963184491992 – 0.8406043323345531

Matmul · 2 workers · reused output

128 × 128 · 128 × 128 · f32

Observed time reduction



CURRENT / BASELINE	95% INTERVAL
0.9857735743644997	0.981228974660111 – 0.9936317176395433

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	169458	169.458
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	196488	196.488
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	99957	99.957
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	86965	86.965
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3440	3.44
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8207	8.207
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3224	3.224
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3504	3.504
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3248	3.248

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21808	21.808
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22543	22.543
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22247	22.247
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22471	22.471
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31111	31.111
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22647	22.647
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	194280	194.28
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168273	168.273
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	208808	208.808
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169098	169.098

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	208280	208.28
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	167786	167.786
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1332074	1332.074
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1475672	1475.672
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1343020	1343.02
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1332212	1332.212
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1330557	1330.557
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1551065	1551.065
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11279	11.279
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	11968	11.968

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12303	12.303
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11272	11.272
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11751	11.751
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20112	20.112
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	85557	85.557
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	93285	93.285
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85788	85.788
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99772	99.772
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	93509	93.509
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87141	87.141

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	665766	665.766
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	728777	728.777
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	767138	767.138
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	757955	757.955
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	678132	678.132
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	766243	766.243
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4416	4.416
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12999	12.999
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3880	3.88
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	18887	18.887

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	92685	92.685
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	697427	697.427
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	50829	50.829
Cholesky factorization / nalgebra 512 × 512 · f64	9204768	9204.768
Cholesky factorization / NumPy 512 × 512 · f64	4027441	4027.441
Cholesky factorization / faer 512 × 512 · f64	2673472	2673.472
Cholesky factorization / Kernix 512 × 512 · f64	4366255	4366.255
Column-pivoted QR / Kernix 96 × 32 · f64	88533	88.533
Column-pivoted QR / faer 96 × 32 · f64	62118.00000000001	62.11800000000001
Column-pivoted QR / nalgebra 96 × 32 · f64	114244	114.244

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	473974	473.974
Column-pivoted QR / nalgebra 256 × 64 · f64	1150714	1150.714
Column-pivoted QR / Kernix 256 × 64 · f64	793153	793.153
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	116964	116.964
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	93901	93.901
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	72397	72.397
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	218537	218.537
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	176609	176.609
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	96173	96.173
Thin QR / Kernix 96 × 32 · f64	72093	72.093

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	65373	65.373
Thin QR / NumPy 96 × 32 · f64	66293	66.293
Thin QR / nalgebra 96 × 32 · f64	61309.00000000001	61.30900000000001
Thin QR / faer 256 × 64 · f64	447455	447.455
Thin QR / Kernix 256 × 64 · f64	695948	695.948
Thin QR / NumPy 256 × 64 · f64	573546	573.546
Thin QR / nalgebra 256 × 64 · f64	586681	586.681
Cholesky factorization / NumPy 64 × 64 · f64	18039	18.039
Cholesky factorization / nalgebra 64 × 64 · f64	20311	20.311
Cholesky factorization / faer 64 × 64 · f64	13255	13.255

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	27703	27.703
Cholesky factorization / Kernix 128 × 128 · f64	280005	280.005
Cholesky factorization / nalgebra 128 × 128 · f64	128682	128.682
Cholesky factorization / faer 128 × 128 · f64	64950.00000000001	64.95000000000001
Cholesky factorization / NumPy 128 × 128 · f64	86549	86.549
Cholesky factorization / faer 256 × 256 · f64	407640	407.64
Cholesky factorization / Kernix 256 × 256 · f64	771122	771.122
Cholesky factorization / nalgebra 256 × 256 · f64	1035745	1035.745
Cholesky factorization / NumPy 256 × 256 · f64	767587	767.587
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	5007	5.007

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8368	8.368
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3496	3.496
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7288	7.288
Model inference / Candle batched normalized/sliced MLP · f64	5287	5.287
Model inference / Kernix batched normalized/sliced MLP · f64	3064	3.064
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3848.0000000000005	3.8480000000000005
Model recording + forward / Candle batched normalized/sliced MLP · f64	7640	7.64
Model backward / Kernix prepared graph → fresh gradients · f64	4175	4.175
Model backward / Candle prepared graph → fresh gradients · f64	21136	21.136
Parameter update / Kernix transactional SGD, fixed gradients · f64	720	0.72

Recorded precision / exact latency table

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WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	46742	46.742
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	16472	16.472
Row softmax / Candle 256 × 64 · f64	110555	110.555
Row softmax / ndarray 256 × 64 · f64	97389	97.389
Row softmax / Kernix 256 × 64 · f64	113236	113.236
Row softmax / NumPy 256 × 64 · f64	133851	133.851
Create zeros / Kernix 64 × 64 · f32	200	0.2
Create zeros / faer 64 × 64 · f32	360	0.36
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / Candle 64 × 64 · f32	232	0.232

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	512	0.512
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1520	1.52
Create ones / Kernix 64 × 64 · f32	248	0.248
Create ones / ndarray 64 × 64 · f32	208	0.208
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997